EX. NO: 4

IMPLEMENTATION OF HILL CIPHER

AIM:

To write a C program to implement the Hill Cipher substitution technique using a 2×2 or 3×3 key matrix.

ALGORITHM:

1. Start the program.
2. Enter the size of the key matrix, either 2 or 3.
3. Enter the key matrix.
4. Enter the plaintext according to the selected matrix size.
5. Convert the plaintext characters into numerical values (`A = 0, B = 1,..., Z = 25`).
6. Multiply the key matrix with the plaintext values.
7. Take the result modulo 26.
8. Convert the numerical values back into letters.
9. Display the encrypted text.
10. Stop the program.

PROGRAM:

```
#include <stdio.h>

#include <ctype.h>



void main()

{

    int key[3][3], text[3], cipher[3];
```

```
int n, i, j;


printf("Enter matrix size (2 or 3): ");

scanf("%d", &n);


printf("Enter the key matrix:\n");


for(i = 0; i < n; i++)

{

    for(j = 0; j < n; j++)

    {

        scanf("%d", &key[i][j]);

    }

}


printf("Enter the plain text (%d letters): ", n);


for(i = 0; i < n; i++)

{

    char ch;
```

```
    scanf(" %c", &ch);

    text[i] = toupper(ch) - 'A';

}



for(i = 0; i < n; i++)

{

    cipher[i] = 0;



    for(j = 0; j < n; j++)

    {

        cipher[i] = cipher[i] + key[i][j] * text[j];

    }



    cipher[i] = cipher[i] % 26;

}



printf("\nPLAIN TEXT: ");



for(i = 0; i < n; i++)

{
```

```
        printf("%c", text[i] + 'A');
    }

    printf("\nENCRYPTED TEXT: ");

    for(i = 0; i < n; i++)
    {
        printf("%c", cipher[i] + 'A');
    }
}
```

OUTPUT:

main.c | Share | Run | Output

```c
    printf("Enter the key matrix:\n");

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &key[i][j]);
        }
    }

    printf("Enter the plain text (%d letters): ", n);

    for(i = 0; i < n; i++)
    {
        char ch;
        scanf(" %c", &ch);
        text[i] = toupper(ch) - 'A';
```

```
Enter matrix size (2 or 3): 3
Enter the key matrix:
3 4 5
2 3 4
1 2 3
Enter the plain text (3 letters): CAT

PLAIN TEXT: CAT
ENCRYPTED TEXT: XCH

=== Code Exited With Errors ===
```

RESULT:

Thus, the Hill Cipher substitution technique was successfully implemented in C language using both 2×2 and 3×3 key matrices, and the given plaintext was successfully encrypted.